

A Weak-Norm Counterexample to Problem 13.1 in Blei’s Grothendieck Inequality Revisited

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Status

This packet gives a likely valid counterexample to Problem 13.1 in Ron Blei’s “The Grothendieck inequality revisited”, arXiv:1111.7304. It proves that both inclusions displayed in (13.1) of the source paper are proper.

Source Problem

The source problem appears in Section 13.1, “Some loose ends”, on PDF pages 92–93. The displayed inclusions are:

13. SOME LOOSE ENDS

13.1. $\tilde{\mathcal{V}}_2(X \times Y)$ **vs.** $\tilde{V}_2(X \times Y)$, $\mathcal{G}_2(X \times Y)$ **vs.** $G_2(X \times Y)$. If X and Y are topological spaces, then

$$\begin{aligned}\tilde{V}_2(X \times Y) &\subset C_b(X \times Y) \cap \tilde{\mathcal{V}}_2(X \times Y), \\ G_2(X \times Y) &\subset C_b(X \times Y) \cap \mathcal{G}_2(X \times Y).\end{aligned}\tag{13.1}$$

The problem statement is:

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Problem 13.1. *Are the inclusions in (13.1) proper inclusions?*

See §3.1 for definitions, and Remark 3.4 for a brief discussion.

13.2. $\mathcal{G}_2(X \times Y) = \tilde{\mathcal{V}}_2(X \times Y)$ **vs.** $G_2(X \times Y) = \tilde{V}_2(X \times Y)$. The first equality is the Grothendieck theorem for discrete spaces X and Y , as per Theorem 2.7 and the second is its “upgraded” version for topological spaces X and Y , as per Corollary 3.6

In text form, Problem 13.1 asks whether, for topological spaces X and Y , the inclusions

$$\tilde{V}_2(X \times Y) \subset C_b(X \times Y) \cap \tilde{V}_2(X \times Y), \quad G_2(X \times Y) \subset C_b(X \times Y) \cap \mathcal{G}_2(X \times Y)$$

are proper.

Proof Intuition

The source's topological classes require Hilbert factorizations whose coordinate maps are norm-continuous after the given topologies on X and Y are taken into account. The set-theoretic classes forget this continuity. The unit ball of an infinite-dimensional Hilbert space supplies exactly the gap: the inner product is jointly continuous for weak topology in the first variable and norm topology in the second, and it has the obvious set-theoretic Hilbert factorization. But any topological G_2 -factorization would force the weak-to-norm identity map on the Hilbert ball to be continuous, which is ruled out by an orthonormal sequence.

Counterexample

Theorem 1. *Let $H = \ell^2(\mathbb{N})$. Let*

$$X = (B_H, \sigma(H, H^*)), \quad Y = (B_H, \|\cdot\|)$$

be the closed unit ball of H , with the weak topology on X and the norm topology on Y . Define

$$f : X \times Y \rightarrow \mathbb{C}, \quad f(x, y) = \langle x, y \rangle.$$

Then

$$f \in C_b(X \times Y) \cap \mathcal{G}_2(X \times Y)$$

but $f \notin G_2(X \times Y)$. Consequently the second inclusion in (13.1) is proper. Moreover, using the discrete Grothendieck theorem $\mathcal{G}_2 = \tilde{V}_2$ and the inclusion $\tilde{V}_2 \subset G_2$ from the source paper, the same f also shows that

$$\tilde{V}_2(X \times Y) \subset C_b(X \times Y) \cap \tilde{V}_2(X \times Y)$$

is proper.

Proof. The function f is bounded by 1. It is jointly continuous on $X \times Y$: if $x_i \rightarrow x$ weakly in B_H and $y_i \rightarrow y$ in norm, then

$$|\langle x_i, y_i \rangle - \langle x, y \rangle| \leq |\langle x_i, y_i - y \rangle| + |\langle x_i - x, y \rangle| \leq \|y_i - y\| + |\langle x_i - x, y \rangle| \rightarrow 0.$$

Thus $f \in C_b(X \times Y)$.

As a set-theoretic kernel, f belongs to $\mathcal{G}_2(X \times Y)$, where the topologies on X and Y are ignored. Indeed, put $\Omega = \mathbb{N}$ and $\mu(\{n\}) = 2^{-n}$. For $x = (x_n), y = (y_n) \in B_H$, define

$$g_x(n) = 2^{n/2} x_n, \quad h_y(n) = 2^{n/2} \overline{y_n}.$$

Then $\|g_x\|_{L^2(\mu)} = \|x\|_2 \leq 1$ and $\|h_y\|_{L^2(\mu)} = \|y\|_2 \leq 1$, while

$$\int_{\Omega} g_x(\omega) h_y(\omega) d\mu(\omega) = \sum_{n=1}^{\infty} x_n \overline{y_n} = f(x, y).$$

Suppose, toward a contradiction, that $f \in G_2(X \times Y)$. By Definition 3.2 of the source paper, there are a probability space (Ω, μ) and $L^2(\mu)$ -continuous maps

$$g : X \rightarrow L^2(\Omega, \mu), \quad h : Y \rightarrow L^2(\Omega, \mu)$$

such that

$$f(x, y) = \int_{\Omega} g(x)(\omega) h(y)(\omega) d\mu(\omega)$$

and

$$C := \sup_{x \in X, y \in Y} \|g(x)\|_2 \|h(y)\|_2 < \infty.$$

Since f is not identically zero, choose $x_0 \in X$ with $g(x_0) \neq 0$. Then

$$M := \sup_{y \in Y} \|h(y)\|_2 \leq C / \|g(x_0)\|_2 < \infty.$$

For all $x, x' \in B_H$,

$$\|x - x'\| = \sup_{y \in B_H} |\langle x - x', y \rangle| = \sup_{y \in B_H} |f(x, y) - f(x', y)| \leq M \|g(x) - g(x')\|_2.$$

The map $g : X \rightarrow L^2(\Omega, \mu)$ is continuous from the weak topology on B_H to the Hilbert norm topology. The displayed estimate therefore implies that the identity map

$$(B_H, \sigma(H, H^*)) \longrightarrow (B_H, \|\cdot\|)$$

is continuous. This is impossible in infinite dimension: if (e_n) is an orthonormal sequence in H , then $e_n \rightarrow 0$ weakly while $\|e_n - 0\| = 1$ for every n .

Hence $f \notin G_2(X \times Y)$. Since the source paper proves $\tilde{V}_2(X \times Y) \subset G_2(X \times Y)$, $f \notin \tilde{V}_2(X \times Y)$. Also, the source's Theorem 2.7 identifies the set-theoretic Grothendieck class $\mathcal{G}_2$ with $\mathcal{V}_2$. Therefore f belongs to the right side of both inclusions in (13.1) and to neither left side. Both inclusions are proper. $\square$

Verification and Novelty Notes

No computation is used. The only functional-analytic input beyond the source definitions is the elementary fact that an orthonormal sequence in an infinite-dimensional Hilbert space converges weakly to 0 but not in norm.

Bounded novelty search performed before promotion:

- Run indexes were searched for ‘1111.7304’, ‘Grothendieck inequality revisited’, ‘Problem 13.1’, ‘proper inclusions’, and close $\tilde{V}_2/G_2$ phrases. No exact prior packet was found.
- Local parsed arXiv sources were searched for the exact Problem 13.1 wording and weak-topology/norm-topology variants. No separate answer was identified.
- Web searches for the exact Problem 13.1 wording, the source title with ‘Problem 13.1’, and close $\tilde{V}_2/G_2$ proper-inclusion phrases returned no direct answer.

Novelty confidence is moderate pending human literature review; the proof is short and self-contained.

Human Review Recommendation

Reviewers should check the alignment of notation with Definitions 2.4, 3.2, Proposition 3.3, Corollary 3.6, and Problem 13.1 of the source paper. In particular, verify that the explicit representation over $(\mathbb{N}, \mu(\{n\}) = 2^{-n})$ satisfies the source’s set-theoretic $\mathcal{G}_2$ definition, while topological G_2 requires L^2 -norm continuous representing families.

References

- [1] Ron Blei, *The Grothendieck inequality revisited*, arXiv:1111.7304.